



Monitoring Resource Utilization

Keeping AI Clusters Efficient and Reliable

A comprehensive approach to visibility, optimization, and reliability for modern AI infrastructure

Why Resource Monitoring Matters

Resource monitoring provides critical visibility into your AI infrastructure's compute, storage, and network usage patterns. This visibility is the foundation for:

Preventing Resource Waste

Identify underutilized GPUs and optimize resource allocation

Efficient Scaling

Make data-driven decisions about infrastructure expansion

Early Bottleneck Detection

Catch performance issues before they impact training jobs



Critical Resources to Monitor



Compute Usage

Track CPU & GPU utilization across training and inference workloads

- Core utilization percentages
- CUDA memory allocation
- Temperature & power draw



Memory Usage

Monitor to prevent OOM crashes that terminate long-running jobs

- System RAM consumption
- VRAM usage patterns
- Memory leaks detection



Storage & I/O

Measure dataset read/write speeds and storage capacity

- Disk IOPS & throughput
- Storage capacity usage
- Cache hit ratios



Network Traffic

Track distributed training communication patterns

- Inter-node bandwidth
- Packet loss rates
- Data transfer latency

Essential Monitoring Tools

Open Source Solutions

Prometheus

Time-series metrics collection at scale with powerful query language

NVIDIA DCGM

Deep GPU health monitoring, utilization tracking, and diagnostics

Grafana

Customizable dashboards for visualizing metrics with alerting capabilities

Cloud Provider Tools



- **AWS CloudWatch** - Native EC2 and SageMaker metrics
- **GCP Monitoring** - Integrated with Vertex AI
- **Azure Monitor** - ML workspace metrics

Benefits of Comprehensive Monitoring

85%

Cost Reduction

Average savings achieved by
identifying and reclaiming
idle GPU resources

3.5x

Training
Throughput

Typical improvement after
addressing bottlenecks
identified through
monitoring

40%

Failure Reduction

Decrease in training job
failures after implementing
proactive monitoring

24hrs

MTTR
Improvement

Reduction in mean time to
resolution when issues are
detected early

Effective monitoring transforms reactive firefighting into proactive optimization, creating a foundation for reliable AI infrastructure that scales predictably.

Common Monitoring Challenges



Metrics Overload

Thousands of available metrics make it difficult to identify the truly important signals from noise



GPU Complexity

GPUs have unique metrics beyond CPU (CUDA cores, tensor cores, memory bandwidth) requiring specialized tools



Fragmented Visibility

Multi-cloud and hybrid environments create monitoring silos that complicate holistic analysis



Alert Fatigue

Poorly tuned thresholds lead to false positives that erode trust in the monitoring system



"The hardest part isn't collecting metrics—it's knowing which ones actually matter for your specific workloads."

— Senior MLOps Engineer

Monitoring Best Practices

1 — Define Clear KPIs

Establish specific targets for resource efficiency:

- GPU utilization consistently >70%
- Memory headroom of 15-20%
- Storage IOPS within 80% of capacity

2 — Implement Tiered Alerting

Create graduated response thresholds:

- **Warning:** >85% memory utilization
- **Critical:** >95% memory utilization

3 — Build Purpose-Built Dashboards

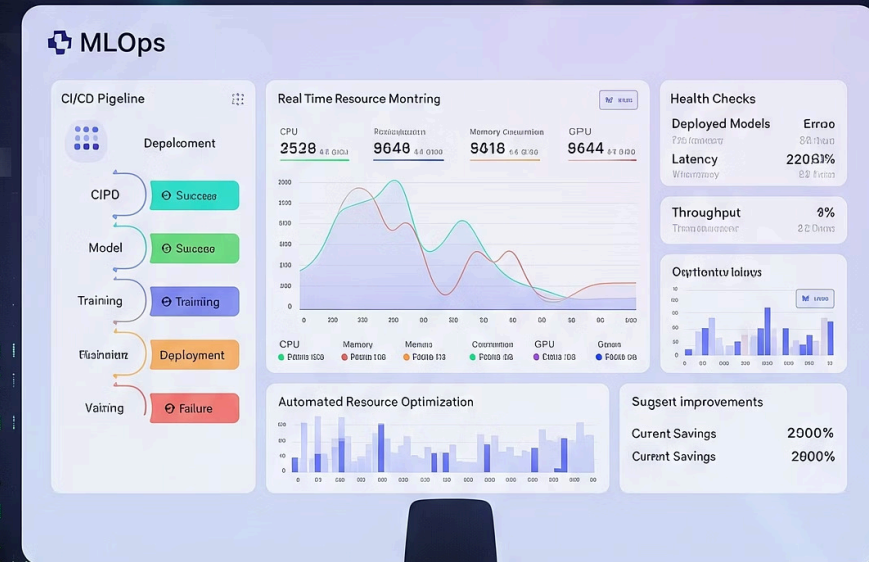
Tailored views for different stakeholders:

- Executive: cost and efficiency metrics
- MLOps: detailed technical indicators
- ML Engineers: job-specific performance

4 — Integrate with MLOps Workflow

Embed monitoring in your CI/CD pipeline:

- Resource checks in pre-deployment tests
- Automated scaling based on utilization
- Performance regression detection



Key Takeaways

Foundation for Success

Resource monitoring is not optional—it's the bedrock of efficient, stable AI infrastructure

Holistic Approach

Monitor the full resource spectrum: compute, memory, storage, and network

Modern Tooling

Leverage purpose-built tools like Prometheus, Grafana, and NVIDIA DCGM

Actionable Insights

Transform raw metrics into concrete infrastructure optimizations

Next Steps: Audit your current monitoring setup, identify gaps, and implement a comprehensive monitoring strategy within the next 30 days.



 Want to discuss implementing these practices in your environment? Contact the MLOps team at mlops@company.com